Arithmetic and complexity

Knuth–Stolarsky lower bound for addition chains

Difficulty: challenging **Importance:** interesting to specialist

Topic: lower bounds for multiplication-only powering

Status: open; admitted after a bounded literature search found no resolution.

Context and notation

An addition chain for a positive integer n is a sequence $1 = a_0 < a_1 < \cdots < a_r = n$ in which every a_i with $i > 0$ is a sum of two earlier terms, which may coincide. Let $\ell(n)$ be its minimum possible length r. The case $n = 1$ uses $\ell(1) = 0$.

Problem statement

Let $\nu(n)$ be the number of ones in the binary expansion of a positive integer n. Is

$$\ell(n) \geq \lfloor \log_2 n \rfloor + \lceil \log_2 \nu(n) \rceil \qquad (n \geq 1)?$$

These rounding conventions are part of the statement. This would constrain multiplication chains for A^n and complements the upper bound in AC-07.

References and status

E. G. Thurber, *The Scholz–Brauer problem on addition chains*, Pacific Journal of Mathematics 49 (1973), p. 229, states the equivalent small-step conjecture and treats a restricted case. H. Altman, *Internal structure of addition chains: Well-ordering* (2018), Conjecture 1.7 and §4, distinguishes proved cases from the general claim. Searches for "Knuth Stolarsky conjecture 2026 proof" found no general resolution. Bläser's Research Problem 2.10 is a discovery lead; this entry uses the explicit floor/ceiling formulation in the specialized sources. **Admitted: no resolution located.**